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CURRENT OPINION

Increased Risk of Musculoskeletal Injury Following Sport-Related Concussion: A Perception–Action Coupling Approach

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Abstract

Recent studies have concluded that athletes have increased risk of musculoskeletal injury following sport-related concussion. While an underlying explanation is still unknown, perceptual-motor control may be implicated in this increased risk. Some authors have purported that indirect perception (i.e., a "top-down" view of neuromuscular control) may be disrupted following sport-related concussion. Direct perception theory states that the athlete and environment are inextricably linked in a continuous perception–action coupling loop. That is, the athlete is able to directly perceive opportunities for action (e.g., "affordances") in the environment. Based on these notions, the aim of the current paper was to introduce a theoretical model that argues that sport-related concussion may dysregulate the direct perception process, potentially increasing behavioral risk of musculoskeletal injury during sport. Our model is integrated with a sport-related concussion clinical treatment model, which highlights individualized profiles that characterize the heterogeneous response to sport-related concussion. These profiles have a typical constellation of symptoms (e.g., anxiety, fatigue, ocular dysfunction, etc.), which themselves have been associated with disrupted perception–action coupling, independent of sport-related concussion. Therefore, we argue that athletes who have not re-established perception–action coupling loops following sport-related concussion may be at increased risk of subsequent musculoskeletal injury.

1 Introduction

Following sport-related concussion (SRC), athletes may have increased risk of musculoskeletal injury (MSKI). In a literature review summarizing findings from ten studies published since 2009, Howell et al. [1] reported increased musculoskeletal risk following SRC across sport and skill level (i.e., college, professional) [1–11]. Based on the included studies, the odds of sustaining a subsequent MSKI range from 1.59 to 3.39 for athletes with recent SRC compared to athletes without SRC over the following year. Similarly, in their meta-analysis of eight studies, McPherson et al. [12] indicated that athletes with a recent SRC were two times more likely to sustain a MSKI compared to athletes without SRC. Howell et al. [1] postulated that neuromuscular control deficits may underlie these deficits, which can persist beyond

Key Points

- Following sport-related concussion, athletes have increased risk of musculoskeletal injury.
- From a direct perception theory viewpoint, this injury risk is related to perceptual motor control, as numerous symptoms and/or impairments following concussion have been associated with disrupted perception–action coupling.
- Restoration of perception–action coupling loops may improve safer return to play outcomes.

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resolution of symptoms and normalization of typical SRC clinical assessments (e.g., balance, cognitive). However, these proposed explanations for the increased risk of MSKI following SRC have yet to be confirmed experimentally.

Howell et al. [1] postulated that divided attention during physical activity with increasing cognitive load, which commonly occurs in sport, could place athletes at increased risk of subsequent MSKI following SRC [1]. This hypothesis is based on evidence that greater deficits are observed in recently concussed subjects during dual-task performance as the cognitive tasks increase in complexity [1, 13, 14]. Perceptual and motor impairments have both been demonstrated in concussed athletes, and it has been suggested that the concussed brain is unable to properly divide cognitive resources when confronted with various stimuli simultaneously [1, 15–18]. This hypothesis follows the assumption that movement is controlled by a "motor executive" in the central nervous system, the role of which is to transform perceptual stimuli into useful internal representations before enacting a motor plan [19]. Under this theory of indirect perception, the motor executive: (1) uses memory to convert otherwise meaningless environmental stimuli into meaningful representations of the world, (2) determines the appropriate response to the task constraints, and (3) enacts a motor program to control the neuromuscular system [20, 21]. To control movement in dynamic environments, such as those encountered during sport, the brain would have to achieve these steps over an extremely limited period of time during the course of play.

Over the past few decades, it has been argued that viewing the sensorimotor system in this hierarchical, "top-down" manner creates computational problems for quantifying coordination of athletes in real-time competition. For example, it has been argued that the number of available degrees of freedom for any athlete during competition is astronomical [22]. Consequently, the limited storage capacity of the brain cannot account for the possible combinations of muscle contractions to execute a nearly infinite number of movements on a very short time scale [23]. Thus, the computational requirements of the indirect perception model suggest that perception and action must be tightly coupled in order to accomplish goal-directed action [19, 24]. Due to the processing and storage capacity problems of indirect perception theory, there is a need for an alternative explanation of the increased MSKI risk following SRC. Here, we propose the direct perception theory, which states that environmental information is directly detected by the perceiver and acted upon, without a need to expand upon the information internally [25]. Therefore, the aim of this Current Opinion paper is to contend that concussion may dysregulate the direct perception process, potentially increasing behavioral risk of MSKI during play, as well as to introduce a novel theoretical model to guide future research investigating this phenomenon. To accomplish this goal, we summarize critical tenets of direct perception theory, discuss the research demonstrating altered direct perception following symptoms commonly associated with SRC, introduce our theoretical model, and discuss opportunities for future research.

2 Direct Perception and Its Theoretical Relationship to Sport-related Concussion

Gibson first introduced direct perception theory, which states that perception is an ongoing activity based upon the notion that the performer (i.e., athlete) and environment are inextricably linked in a continuous perception–action coupling loop [23]. Gibson expanded on this theory by introducing the concept of affordances, or possibilities for action for a given performer in a specific environment [26]. Actualizing an affordance—that is, turning one's perceived opportunity for action into bodily motion—is achieved through performatory movement, as opposed to exploratory movement [25]. For example, a soccer player who makes a clever pass (i.e., performatory movement) may have been informed of his or her opportunities for action by rapidly rotating their head with the goal of perceiving a wider range of environmental dynamics (i.e., exploratory movement) [27, 28]. Exploratory movement, which may involve head/neck rotation or movement into open spaces, is still critical as it provides the link between perception and action [25, 26]. Exploring one's environment is critical for detecting affordances and providing information necessary for prospective action during performatory movement [27, 28].

Detection of affordances is strongly related to the perceiver's physical capabilities—also termed action capabilities; as such, affordances are only available to those with the physical ability to actualize them [19, 23]. Consider an example of two basketball players whose goal is to score, where one is significantly taller than the other. In this situation, a basketball may afford "slam-dunking" for the taller individual but "laying-up" for the smaller, based on differences in jumping ability or body height. In a situation where both players are fully capable of both actions (i.e., their heights and jumping abilities are equal), the most inviting affordance will emerge during the context of play [29]. Thus, an athlete's individual kinematic and kinetic characteristics make up their action capabilities [30]. These capabilities are constantly recalibrated, based on interacting with and exploring the environment as well as the changing state of the athlete [30, 31]. For example, a soccer player attempting to head a corner kick towards the goal must be calibrated to their current jumping height and approach speed capabilities in order to perceive whether heading is afforded. The limits of one's action capabilities are referred to as action boundaries [32]. Misestimation of one's action capabilities

and associated action boundaries in relation to situational information may contribute to increased behavioral risk (i.e., increasing risk of MSKI based on movement behaviors) [33, 34]. Further, athletes need to continually tune-into perceptual information that guides future movement. Therefore, attunement is achieved when an athlete attends to the proper perceptual constancies, or invariants, which emerge from repeated action within the environment [23, 35]. To fully appreciate the relevance of calibration, attunement, and exploration, it is important to emphasize that movement is guided prospectively; the athlete requires the information about action capabilities and future opportunities for action prospectively, i.e., "ahead of time" [27, 28, 36]. The ongoing processes of calibration, attunement, and exploration are distinct, as calibration relates an athlete's physical abilities to the environment, and exploration and attunement relate to perception of constancies, or invariants, and affordances that emerge from repeated action within the environment [28, 31, 35, 37, 38]. However, all concepts are critical for prospective guidance of movement in the direct perception paradigm.

The processes of recalibration and attunement are refined through experience, as in practice for athletes. Over time, athletes hone tuning parameters to the spatiotemporal constraints of the sport environment, through both exploratory and performatory movements [23, 27, 28, 36]. These processes are highly sensitive in order to enhance the ability of the athlete to adapt to changing internal and external states, which necessitates continuous calibration. However, this sensitivity makes the perceptual motor system highly susceptible to perturbations that could theoretically disrupt the perception–action coupling that had previously been developed and refined through practice and experience. Therefore, we view SRC as a perturbation to the perception–action coupling process due to the myriad of symptoms that commonly result from SRC, such as anxiety, fatigue, and ocular dysfunction, which themselves have been demonstrated to disrupt perception–action coupling [39, 40]. More specifically, we hypothesize that SRC may disrupt the (re)calibration and attunement process, which would theoretically increase behavioral risk through improper temporal execution of movements and/or incorrect body positioning during competition, thereby increasing risk of MSKI (Fig. 1).

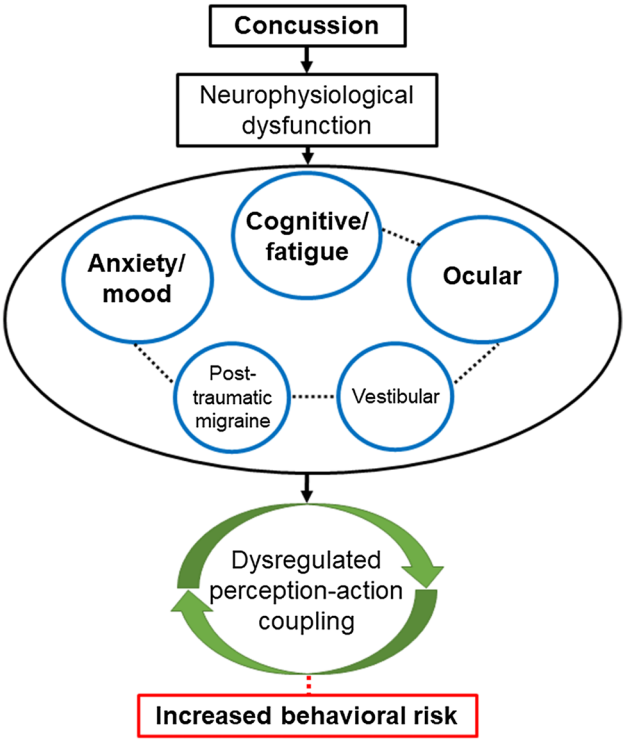

Fig. 1 Theoretical model depicting the relationship between concussion and dysregulation of perception–action coupling, which would potentially increase behavioral risk. Blue circles indicate concussion clinical profiles as described by Collins and Kontos [44]. Dotted black lines indicate clinical associations between profiles. Anxiety/mood, cognitive/fatigue, and ocular have been highlighted due to their documented association with disrupted perception–action coupling. Dotted red lines highlight the hypothesis that neurophysiological dysfunction dysregulates the perception–action coupling loop, which increases behavioral risk during competition

3 Concussion Assessment: A Clinical Profile Approach

Assessing and treating concussion, defined as a traumatic brain injury induced by biomechanical forces, is a complex process that has, historically, been treated with a "one-size-fits-all" approach [41]. In other words, concussions were diagnosed and treated homogeneously across all individuals. However, there is a growing appreciation in the literature that concussion is a heterogeneous injury, both in presentation and the individual's response to treatment [41]. In response to this, Collins et al. [42] have advocated for a comprehensive, interdisciplinary assessment of suspected concussion, including symptom assessment, neurocognitive performance, vestibular/ocular motor disruption, and biopsychosocial history. This comprehensive assessment pairs with a targeted treatment approach, which emphasizes individualized responses to concussion and represents different domains of symptoms/impairment [43]. Further, Collins and Kontos [44] proposed a treatment model in which typical concussion symptoms and impairment represent one or more of six concussion clinical profiles including anxiety/mood, cognitive/fatigue, migraine, ocular, vestibular, and cervical (i.e., neck). Anxiety/mood and cognitive/fatigue are paired together because of the consistent symptom overlap between impairments. Each of these profiles has been significantly associated with a secondary profile in concussed athletes [45]. For example, an athlete with primary vestibular symptoms/impairment is more likely to

present with secondary ocular symptoms/impairment [45]. Categorizing athletes by these profiles enhances rehabilitation by providing a framework for targeted intervention to improve symptoms/impairment and facilitate faster return-to-play [44]. Researchers have proposed alternative clinical treatment models for concussion, such as Ellis et al. [46], who advocate for three post-concussion disorders: physiological, vestibulo-ocular, and cervicogenic. Catastrophizing and pain avoidance has also been related to persistent symptoms following concussion, and suggested to be a profile in itself [47]. However, the authors of the current work have utilized the Collins and Kontos clinical treatment model within our theoretical model, as it encapsulates the heterogeneous response to concussion supported by clinical research and prioritizes an early intervention approach believed to be critical for concussion rehabilitation [44]. In the next section, we discuss each of the clinical profiles, their typical symptoms, and their relationship to disrupted perception–action coupling.

3.1 Anxiety/Mood

The anxiety/mood profile is characterized by common symptoms of anxiety and/or mood disorders, such as excessive worry, difficulty turning off thoughts, rumination, pre-occupation with symptoms, sadness, limited attention and social interactions, and panic attacks [42]. The anxiety/mood profile is a primary driver in 24% of individuals with recent concussion, and is significantly associated with migraine as a secondary profile [45]. Experimental increases in anxiety have been directly associated with disruption of perception–action coupling. For example, Pijpers et al. [48] used indoor wall-climbing traverses at high and low heights to manipulate anxiety and examine the effect on perceiving and realizing affordances. In a series of experiments, the authors found that anxiety reduced perceived maximal reaching height [49]. Daviaux et al. [39] and Graydon et al. [50] also induced anxiety in subjects with the aim of investigating any subsequent changes in action capabilities. Anxiety was achieved by restricting breathing during a seated reach-and-grasp task. In both cases, subjects whose anxiety had increased underestimated their action capabilities. Anxiety can impair attention, which Gibson defined as the process by which affordance detection is controlled [23, 26, 51]. Further, registration of the many informational invariants that guide performance is governed by attention [23]. Thus, it could be hypothesized that anxiety impairs perception–action coupling by interfering with attention, and, therefore, affordance detection.

3.2 Cognitive/Fatigue

The cognitive/fatigue profile is characterized by intense fatigue induced by cognitive activity and/or difficulty with cognitive activity [42]. A retrospective chart review of 236 recently concussed individuals found that cognitive/fatigue is the primary profile in ~ 11% of individuals [45]. An athlete with this profile will report difficulties with concentration, memory, and sleep disruption, leading to low energy and excessive daytime sleepiness [42]. Sleep abnormalities, such as insomnia, excessive daytime somnolence, and circadian disruption, are prevalent following 30–80% of concussions [52]. Moreover, sleep deprivation has been negatively associated with the perception and actualization of action boundaries [40, 53]. Daviaux et al. [40] asked a group undergoing 24 h of sleep deprivation and a habitual sleep group to estimate the maximum height at which they could step over a horizontal bar before and after a night of their respective sleep condition. Actual maximum "crossable" height was also determined for both groups, who were blinded to the result. The sleep-deprivation group underestimated the maximal height that was crossable, with no change in actual performance compared to baseline measurements [40]. Performance on both task components did not change across days for the habitual sleep group. The authors concluded that prolonged wakefulness altered the perception of action capabilities. Specifically, it could be hypothesized that the cognitive/fatigue profile interferes with perception–action coupling by impairing affordance selection, as altered by fatigue and potential sleep disruption.

3.3 Migraine

The migraine profile is characterized by intermittent headache, headache upon wakening, headache with nausea, headache with phono- or photosensitivity, visual aura, motion sickness, and sleep problems [42]. The migraine profile is a primary driver in 26% of individuals, and is significantly associated with vestibular and anxiety/mood secondary profiles [45]. While affordance perception has not been directly studied in those with migraines, the significant relationship between the migraine profile and the vestibular/anxiety profiles suggests an indirect relationship may exist between those with the migraine profile and impaired affordance perception and actualization [45]. Therefore, we hypothesize that those with a primary migraine profile may have impaired perception–action coupling by interfering with the recalibration process or attention.

3.4 Ocular

The ocular profile is characterized by symptoms of blurry vision, difficulty focusing, eye strain, difficulty reading, and headache/fatigue trigged by visual activity [42]. The ocular profile is the primary driver in 16% of individuals with recent concussion and is significantly associated with a secondary cognitive/fatigue profile [45]. Because of the importance of visual perception for guiding movement, we hypothesize any impairment that causes ocular symptoms (e.g., headache/fatigue) to arise as a result of ocular movement will likely impair the perception–action coupling process [54]. Additionally, vision is a critical contributor to the calibration process, as well as the detection of invariants. Vision is also critical for perceptual guidance of locomotion, and may interfere with the process of exploration to guide future movement [31, 55]. Considering that the ocular profile is associated with post-traumatic oculomotor impairment, targeted rehabilitation to allow an athlete with this profile to explore, recalibrate, and reattune will be critical [56].

3.5 Vestibular

The vestibular profile is characterized by dysfunction of the vestibular system, which maintains balance and postural stability in the healthy individual, as well as sensing their movement through the environment [57]. This dysfunction manifests in the recently concussed individual as dizziness, nausea, mental fogginess, anxiety in busy environments, motion sensitivity, and balance problems [42]. The vestibular profile has been shown to be the primary driver in 19% of individuals, with significant associations with ocular and migraine secondary profiles [45]. Vestibular dysfunction can disrupt the understanding of one's body, which may be particularly useful for detection of body-scaled affordances, such as the passability of a narrow aperture [58, 59]. Vestibular dysfunction can also lead to feelings of apprehension in engaging with their environment, due to external stimulation activating uncomfortable symptoms [45]. We hypothesize that a lack of willingness to practice and/or play, due to exacerbation of vestibular symptoms, may significantly impact one's ability to successfully recalibrate by not exploring one's environment [27, 28].

3.6 Summary

The typical characteristics of these concussion clinical profiles dysregulate the perception–action coupling process (Fig. 1) [30, 39, 40, 48]. The authors contend that the individualized neurophysiological response to SRC leads to symptoms and/or impairments that require the individual to recalibrate and/or re-attune to their altered perceptual system to successfully control movement. Moreover, some symptoms or impairments following SRC are often undiagnosed, and therefore untreated, such as ocular and/or vestibular dysfunction [60]. This enhances the likelihood of the athlete returning to play with unresolved physiological deficits that alter the perceptual landscape for that individual. Without adequate time to recalibrate following clearance for return to play, successful performance may be impaired and risk of musculoskeletal injury increased [34].

4 Disrupted Perception–Action Coupling Leads to Increased Behavioral Risk: Practical Applications and Future Research

Researchers have previously approached musculoskeletal injury risk problems from an ecological perspective. Cordovil et al. [34] have suggested that unintentional injuries in children occur when the result of an action has a different outcome than expected. This may occur due to erroneous perceptions of the individual's action capabilities specific to the current environmental situation or an unanticipated actualization of affordances (the coupling of action capabilities to the environment) [33, 34]. The authors also suggest that children who practice less (e.g., have fewer opportunities to develop perceptual attunement at a given task) may experience a higher probability of injury [34]. Further, more active children may be more to prone to injuries resulting from intentionally risky behaviors, as they explore the environment and the limits of their action boundaries [34]. Critically, Plumert et al. [33] have reported that a child's overestimation of their action capabilities in laboratory settings is associated with increased injury risk in everyday life. This ecological model emphasizes that a child's development requires constant recalibration and re-attunement due to consistent changes in physical growth and action capabilities [34, 54]. The authors of the current paper suggest that SRC may temporarily disrupt the perception action coupling process, as a result of the specific symptoms/impairment that occur after the injury, requiring adequate practice to recalibrate/re-attune before returning to play.

The hypothesis that inadequate perceptual-motor attunement is related to MSKI risk in athletes has not been directly investigated. However, two recent studies have clear applications with respect to MSKI risk during athletic competition. Brault et al. [61] investigated novice and experienced rugby players' ability to detect deceptive movement in rugby. Expert players were able to tune into real information about the offensive player's intended movement, rather than deceptive information, in order to intercept the opposing player [61]. Further, Watson et al. [62] analyzed the ability of rugby players to judge the "passability" of a gap between

two defenders (see Fig. 2 for example of gap passage in athletic competition). With practice, participants successfully tuned-in to perceptual-motor information to prospectively judge the passability of the gap. As stated earlier, even when healthy, athletes continually recalibrate to changes within themselves and changes in available environmental information [28, 31]. Because the recalibration process is highly sensitive, one could postulate that an injurious perturbation of the perceptual-motor system, such as SRC, may require substantial exploration and recalibration processes.

It is possible that the increased risk of sustaining another SRC, or MSKI, following a first SRC is related to an inability to re-attune one's bodily system to the dynamics of their specific environment [2–6]. That is, SRC may perturb the perception of variables critical to performance, which would theoretically increase injury risk by simply being in the wrong place at the wrong time [63]. Yet, the current return to play guidelines advocate a six-stage clearance model which does not assess recalibration/re-attunement (Table 1) [64]. Specifically, once stage 5 of the current return-to-play guidelines is reached, only one full-contact practice and 24 symptom-free hours (i.e., athlete does not report headache, nausea, fogginess, dizziness, etc. specifically related to SRC) are required before return to play [64]. While activity in the absence of symptoms is important, motor control deficits have been repeatedly demonstrated following symptom resolution and return-to-play [14, 65–68]. Unfortunately, these motor deficits are subtle and only currently detectable in the research laboratory and not the clinic, such as in dual-task gait testing paradigms [1]. Moreover, athletes returning from SRC may be in

Table 1 Graduated return-to-play strategy, recreated from current sport-related concussion consensus statement [61]

Stage	Activity
1	No activity
2	Light aerobic exercise
3	Sport-specific exercise
4	Non-contact training drills
5	Full contact practice
6	Return to play

A minimum of 24 h without symptom provocation is recommended following exposure to each stage before progressing to the next stage

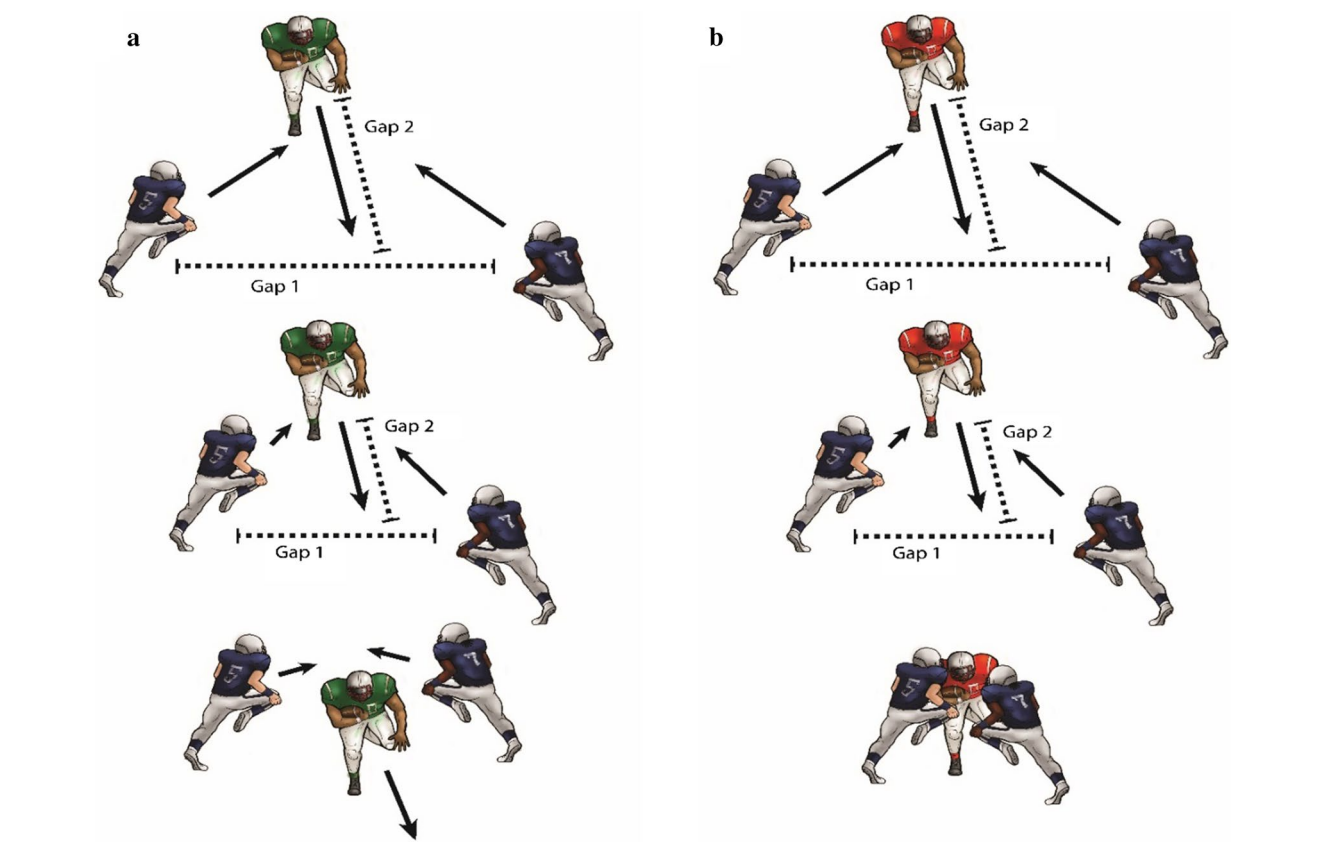

Fig. 2 Simplified example of **a** successful vs. **b** unsuccessful gap passage in American Football. The runner's goal is to avoid being tackled, while the goal of the defenders is to tackle the runner. The runner must perceive the "passability" of the closing gaps in order to achieve the goal. Successful gap passage relates to proper perception of one's action capabilities as well as the spatiotemporal aspects of the closing gaps

a dysregulated perception–action coupling state and therefore unable to recalibrate and/or re-attune to the dynamics of play as quickly as healthy athletes. Ensuring adequate recalibration has occurred during the return-to-play progression could be critical in reducing risk of subsequent musculoskeletal injury, and incorporation of a sensitive, portable evaluation of perception–action coupling behavior could inform clinical decision making on return-to-play readiness.

In response to this need, our preliminary work has developed and utilized the Perception Action Coupling Task (PACT), which is a reliable measure of action boundary perception [69]. PACT revealed persistent deficits in affordance actualization and perception accuracy between controls and athletes with a previous SRC (average time since injury = 264 days) [70]. In introducing this theoretical model, it is the intent of the authors to provide a framework to generate future research. While preliminary evidence shows a relationship between SRC and action boundary perception deficits, it will be critical to establish if these deficits are a precursor to sustaining SRC or occur after SRC (i.e., if action boundary perception deficits are a risk factor for SRC or a resulting impairment). This information can be used clinically to guide how affordance perception deficits in athletes are interpreted. Further, tools like the PACT can provide a clinic-friendly evaluation that quantifies perception–action coupling behavior and may inform clinicians on return-to-play decisions beyond the scope of the SRC evaluation tools that are currently available.

Future research should also investigate which concussion clinical profiles may be more susceptible to affordance perception deficits. For example, we hypothesize that anxiety/mood, cognitive/fatigue, and ocular profiles may be more related to affordance perception deficits due to the established research linking typical symptoms of those profiles to disrupted perception–action coupling [30, 39, 40]. Demonstrating the hypothesized association between affordance perception deficits in laboratory tests and injury risk when returning to athletic competition will also be critical. Finally, development and validation of ecological tests, which can be conducted clinically (such as sport-specific scenarios conducted within a virtual or augmented reality), may help clinicians quantify the recalibration/re-attunement process following SRC and better inform clinical decision making for return to play.

5 Conclusion

Following SRC, athletes' neurophysiologic characteristics such as perception–action coupling appear to be altered [71–74]. We recently reported that this disruption in perception–action coupling is evident in previously concussed athletes who are clinically recovered [70]. Moreover, many symptoms that are common following SRC, such as anxiety, fatigue, and ocular dysfunction, have been associated with altered perception–action coupling [39, 40, 48, 49, 75]. Therefore, we argue that a recently concussed athlete requires adequate time to re-establish perception–action coupling loops, as the perceptual-motor relationship between the athlete and the environment has changed [34]. Athletes who have not re-established perception–action coupling following SRC may be at increased risk of subsequent MSKI. Re-establishing a tightly coupled athlete-environment system may limit the risk of MSKI. Incorporating sensitive and clinic-friendly metrics of perception–action coupling behavior, such as the PACT, into return-to-play decision making may better inform safer clinical decision making.

Compliance with Ethical Standards

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Conflict of interest Shawn Eagle, Anthony Kontos, Gert-Jan Pepping, Caleb Johnson, Aaron Sinnott, Alice LaGoy, and Chris Connaboy declare that they have no conflicts of interest relevant to the content of this article.

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